

```

"""
ARKHEIA-CPS – FastAPI Application Entry
Point
ARKHEIA Contract Protection System
FIA → OIA-PEM → RTCSA pipeline for AUTO
and HOUSING contracts
"""

from fastapi import FastAPI
from fastapi.middleware.cors import
CORSMiddleware

from app.config import get_settings
from app.db import create_all_tables
from app.routers import contracts, analysis

settings = get_settings()

app = FastAPI(
    title="ARKHEIA Contract Protection
System (CPS)",
    description=(
        "Discipline-based real-time
contract safety analysis. "
        "FIA (Identity) → OIA-PEM
(Pattern) → RTCSA (Safety). "
        "Verticals: AUTO (with Perfection
Law) + HOUSING."
    ),
    version="1.0.0",
    docs_url="/docs",

```

```
        redoc_url="/redoc",
    )

# — CORS — allow Webflow and any future
frontends —————
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],      # Tighten to
specific domains in production
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

# — Routers
—————

app.include_router(contracts.router)
app.include_router(analysis.router)

# — Startup
—————

@app.on_event("startup")
def on_startup():
    if settings.ENV == "dev":
        create_all_tables()

# — Health
—————

@app.get("/health", tags=["System"])
```

```
def health():
    return {
        "status": "ok",
        "system": settings.APP_NAME,
        "version": settings.VERSION,
        "env": settings.ENV,
    }

@app.get("/", tags=["System"])
def root():
    return {
        "system": "ARKHEIA Contract
Protection System",
        "version": "1.0.0",
        "endpoints": {
            "submit_auto": "POST
/contracts/auto",
            "submit_housing": "POST
/contracts/housing",
            "get_analysis": "GET
/analysis/{contract_id}",
            "list_analyses": "GET
/analysis",
            "stats": "GET
/analysis/summary/stats",
            "docs": "GET /docs",
        },
    }
```